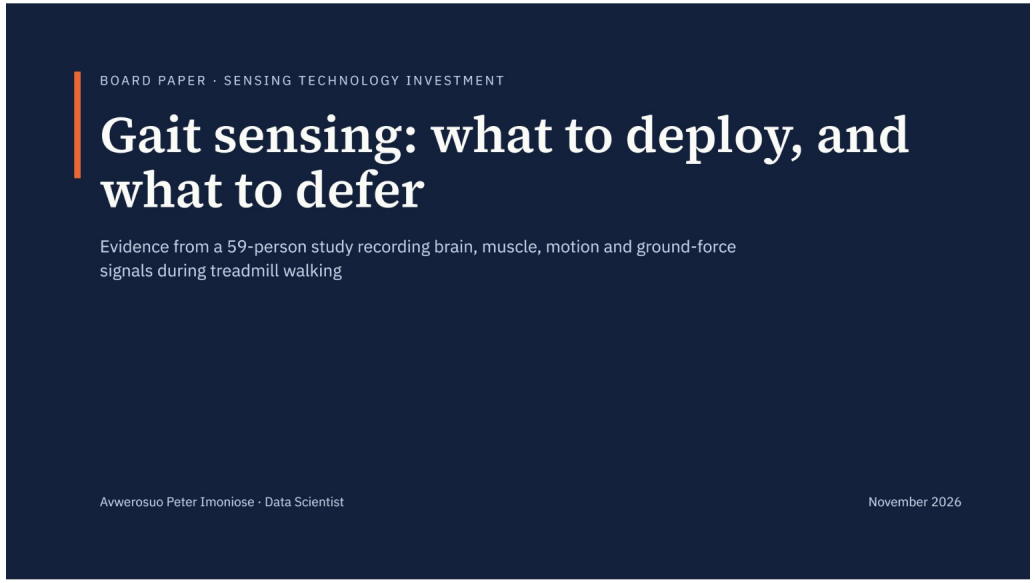
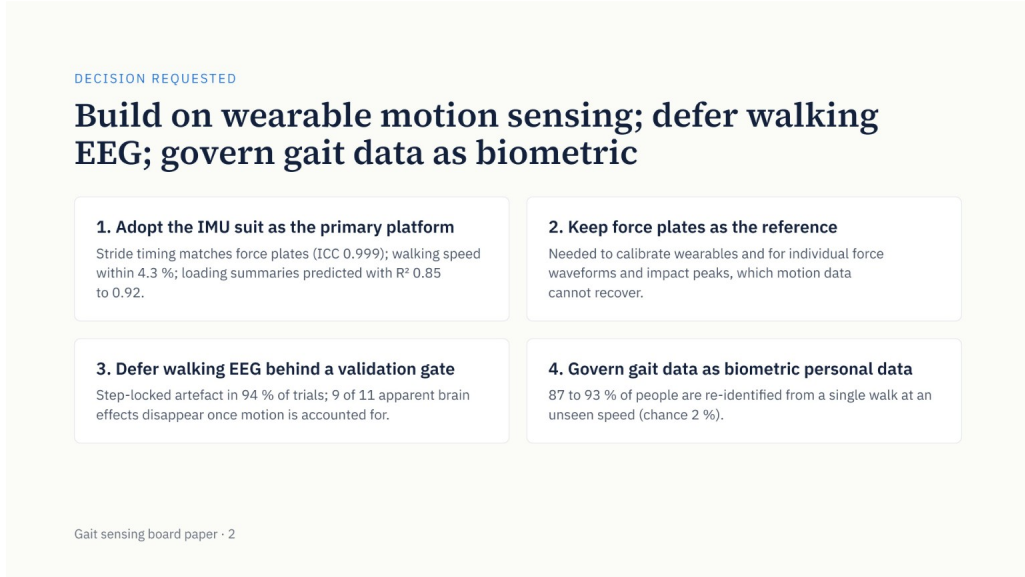


This paper asks the board to decide which gait-sensing technologies to build on. The evidence comes from an independent re-analysis of a public dataset in which 59 healthy adults walked at three speeds while four sensor systems recorded simultaneously. Every number in the deck is reproducible from the accompanying code and notebooks.



Four decisions. Blue cards are investments the evidence supports now; orange cards are constraints: what to hold back and how to handle the data. The rest of the deck gives the evidence for each, in that order.

THE EVIDENCE BASE

Four sensor systems recording the same steps

59
healthy adults, 37 women and 22 men, median age 22

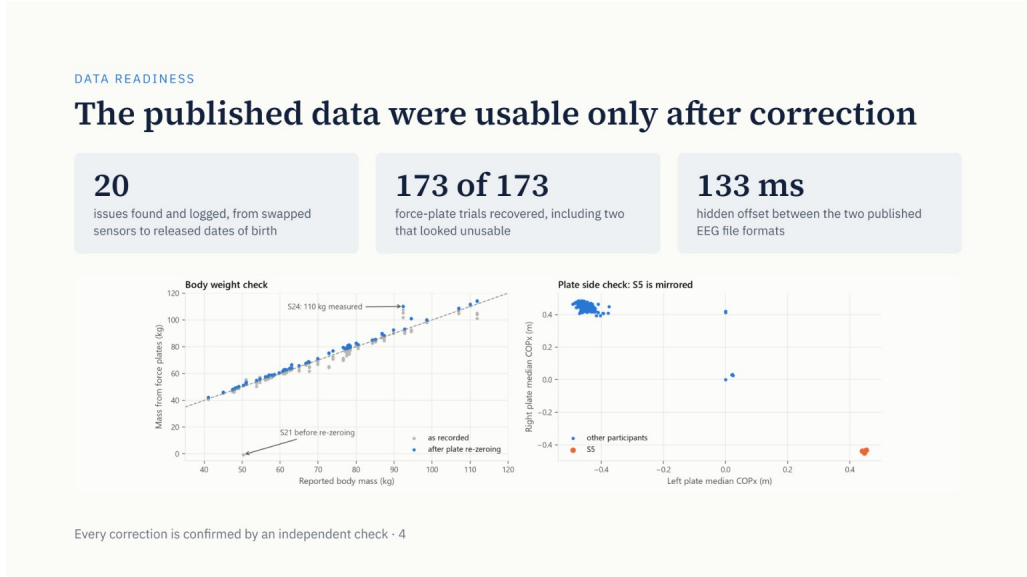
3
treadmill speeds: 0.50, 0.75 and 1.00 m/s, one minute each

15,574
strides measured on the instrumented treadmill

System	What it measures	Role in this paper
Force plates (split belt)	Ground forces under each foot, 1000 Hz	Reference standard
IMU suit (17 sensors)	Body segment motion, 60 Hz	Candidate scalable platform
EMG (12 muscles)	Muscle activation envelope, 16 Hz	Candidate add-on module
Dry EEG (19 channels)	Scalp electrical activity, 300 Hz	Candidate brain-signal module

Source: Katmah et al., Khalifa University, public dataset v1.0.0 (CC BY 4.0) · independent re-analysis · 3

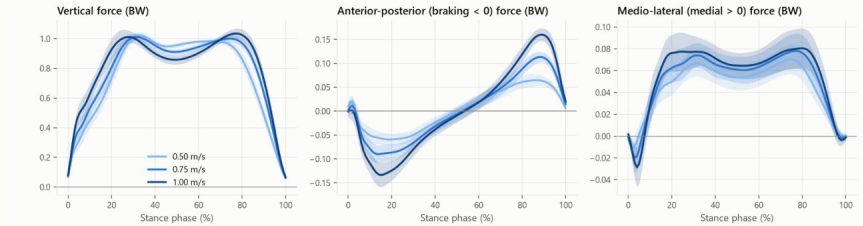
Because all four systems recorded the same steps at the same time, each candidate technology can be judged directly against the force-plate reference on identical walking, which is rarely possible. Limitation to keep in mind throughout: healthy young adults on a treadmill.



Left chart: before correction several participants appear lighter than they are, and one appears weightless, because force plates were not zeroed. After re-zeroing, measured and reported weight agree within about 2 %, except one participant whose recorded mass is wrong in the metadata. Right chart: one participant's left and right plates are swapped. Lesson for the programme: any external dataset or vendor data needs an audit before it informs a decision.

THE REFERENCE MEASUREMENT

Walking faster reshapes loading more than it raises the peak



+50 % stride length, +32 % cadence from 0.5 to 1.0 m/s; longer strides carry most of the gain.

Braking, propulsion and loading rate double; the impact peak stays near 1.03 body weights.

Higher BMI walks more cautiously: longer stance and slower loading per kilogram.

Force plates, 59 participants, mixed models with Holm correction; shaded bands = between-person SD · 5

This is the yardstick the wearable has to meet. The useful message for a product is that the clinically interesting change with speed is in the shape and timing of loading, not in a single peak value, which matters for what a wearable needs to estimate.

FINDING 1 · WEARABLE MOTION SENSING

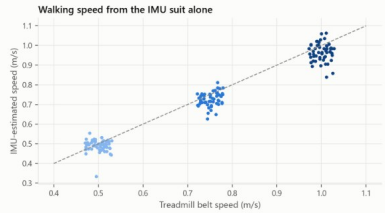
The IMU suit matches the treadmill on timing and speed

0.999
agreement (ICC) on stride time; limits ±12 ms

99 %
of heel strikes and toe-offs detected

4.3 %
mean error in walking speed, with no treadmill input

0.77
agreement on stance %: good for trends, not individual thresholds



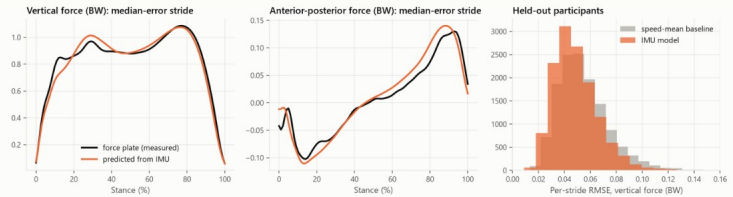
45 held-out participants, 129 trials; the detector was tuned on 10 other participants only · 6

The setting of the event detector was chosen on ten development participants and frozen before these results were computed, so these are honest out-of-sample figures. The orange tile is the caveat: stance percentage is close on average but individual readings can differ by up to about four percentage points.

FINDING 2 · A VIRTUAL FORCE PLATE

Motion data predict loading summaries, not each person's force curve

R^2 0.92	0.84-0.85	0.70	below 0.2
propulsive force peak	braking peak, stance and double support	vertical loading rate	impact peak and side force: not recoverable

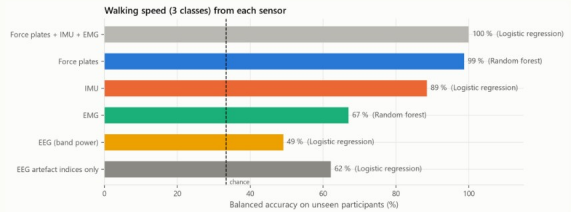


58 participants held out in turn; stride curves fit with r = 0.99 but only 6-10 % below a speed-matched template · 7

Trial-level summaries are well predicted on people the model never saw. The waveform result is the honest limit: the predicted curve looks right, but almost all of that comes from knowing the walking speed, and the motion data add little about the individual. Where a decision depends on a person's own force curve or impact peak, the force plate is still required.

FINDING 3 · COMPARING THE SENSORS

Walking state is carried by the biomechanics, not the brain signal



IMU reaches 89 % on people it never saw, close to the force plates' 99 %.

Combining sensors adds little beyond the force plates.

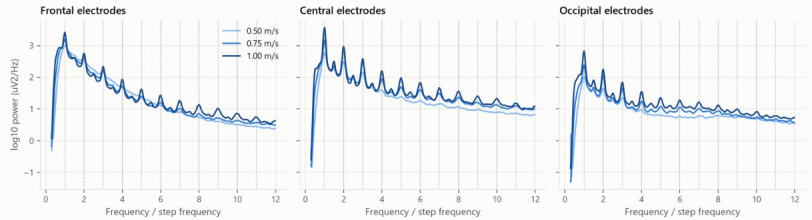
EEG is beaten by its own artefact measures: 62 % against 49 %.

Task: classify 0.50 / 0.75 / 1.00 m/s; 160 trials, 57 participants, 10-fold cross-validation by participant · 8

Speed classification is a probe, not a product: it asks which signals carry information about how a person is walking that transfers to new people. The grey bar at the bottom is the key control: a model given only two measures of motion contamination outperforms the full EEG feature set.

FINDING 4 · BRAIN SIGNAL DURING WALKING

The walking EEG records the footsteps, not the brain



94 %
of trials show peaks at the step rhythm

4×
more coupled to own head motion than to others'

×1.9
heel-strike transient per +0.5 m/s

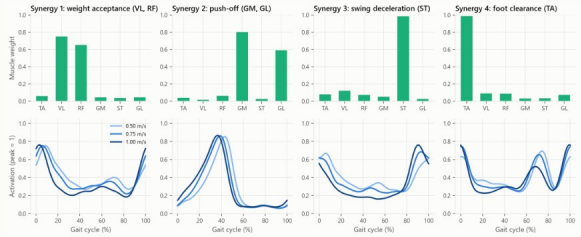
9 of 11
apparent brain effects vanish after motion adjustment

175 trials, 19-channel dry headset; frequency shown as a multiple of each person's step frequency · 9

Each person's spectrum is rescaled so that their own step frequency sits at 1. Brain rhythms have no reason to line up at 1, 2, 3 and so on; mechanical artefact from electrode and cable movement does. Four independent tests agree. This does not say walking EEG is impossible, only that this hardware and this data cannot support claims about the brain without purpose-built artefact removal.

FINDING 5 · MUSCLE ACTIVITY

Muscle coordination follows four shared patterns in everyone



0.98 similarity between each person's patterns and the shared set, at every speed.

Effort scales with speed: main muscles work 40-49 % harder per +0.5 m/s.

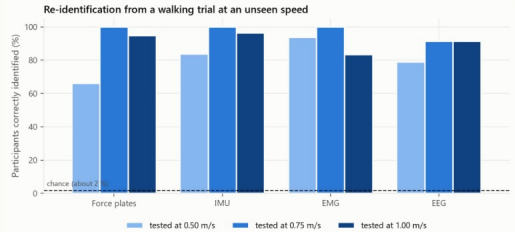
Limit: only a 16 Hz envelope was released; raw signal is needed for more.

289 leg-trials from 56 participants; non-negative matrix factorisation · 10

Stable, shared coordination patterns are what make muscle-level monitoring plausible: a deviation from the shared pattern would be interpretable. EMG is weak on its own as a classifier, so it is positioned as an optional module that adds explanation, captured as raw signal in any future study.

FINDING 6 · PRIVACY

A walk identifies the walker: gait data are biometric



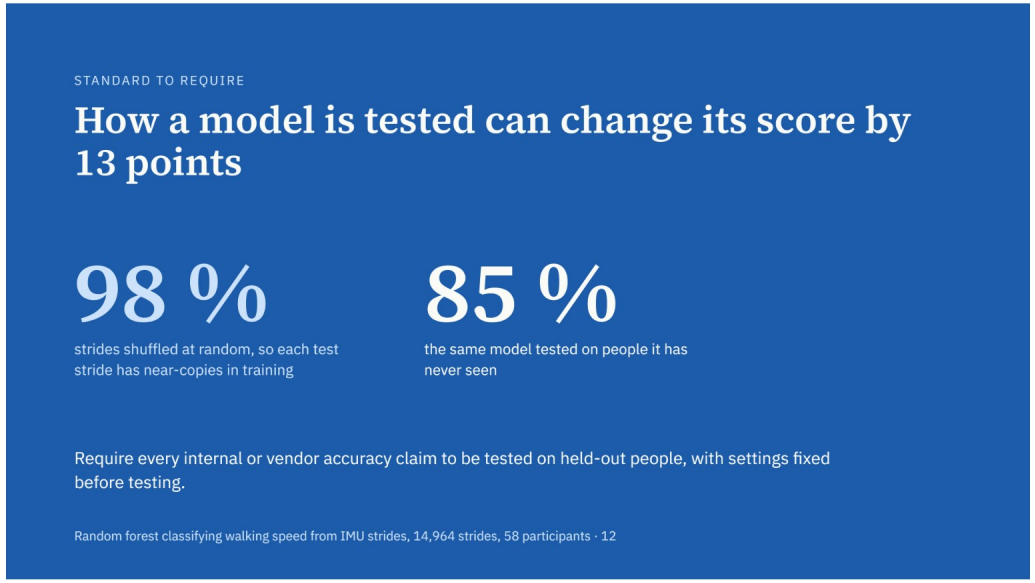
87-93 % re-identified from one walk at an unseen speed; chance is 2 %.

88 % correct on sex inferred from gait alone, against a 62 % baseline.

The “anonymised” release still contained 59 full dates of birth.

Train on two speeds, test on the third; one recording session, so recognition across days is not tested · 11

Removing names does not anonymise gait data. Every sensor, including the cheapest, recognises people across walking conditions. The recommendation is to treat all collected gait data as biometric personal data from day one: a data protection impact assessment, access control, and no release of raw signals without a re-identification test.



Same data, same model, two testing protocols. The shuffled split lets strides from the same minute of walking appear in both training and testing, so the model is partly recognising the person rather than the walking speed. Deployment means new people, so only the second number is meaningful. This is the single cheapest safeguard for the board's future decisions.

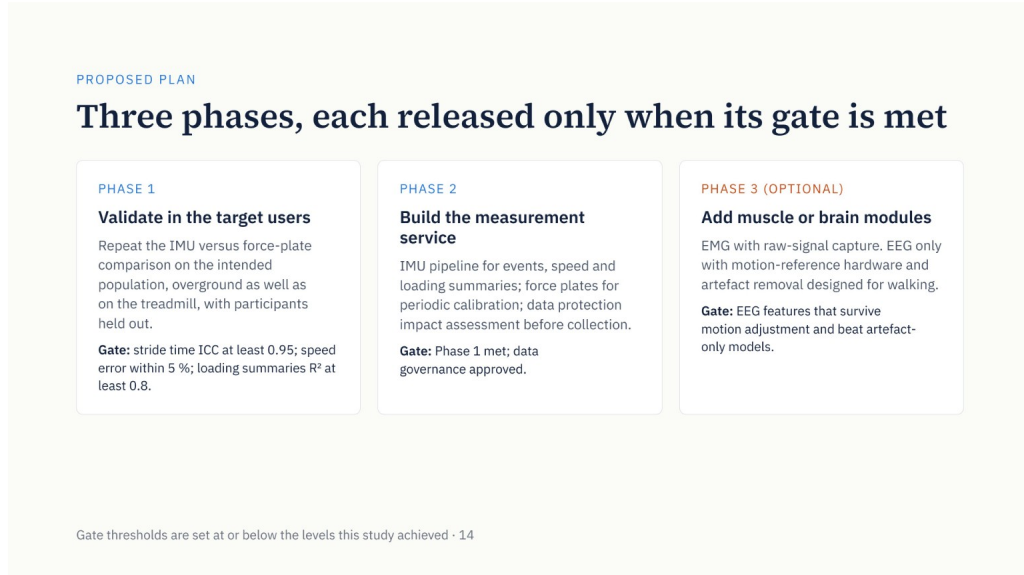
RECOMMENDATION

Each technology has a clear role, or a clear reason to wait

Technology	Supported by the evidence	Not supported	Decision
IMU suit	Gait events, stride time, walking speed, loading summaries	Individual force waveforms, impact peak	Adopt as primary platform
Force plates	Reference timing and forces (speed classified at 99 %)	Portability, use outside the lab	Retain for calibration
EMG	Muscle timing, effort scaling, stable coordination patterns	Stand-alone use (speed at 67 %); raw-signal analysis	Optional module, raw capture
Dry EEG	Nothing beyond its motion-artefact content	Any claim about brain activity during walking	Defer behind a validation gate

Accuracy figures are for people the models had never seen, which is the deployment condition.

The decision matrix restates slides 6 to 10 as choices. Adopt and Retain are investment decisions; Optional and Defer are scope decisions that avoid spend on capability the evidence does not yet support.



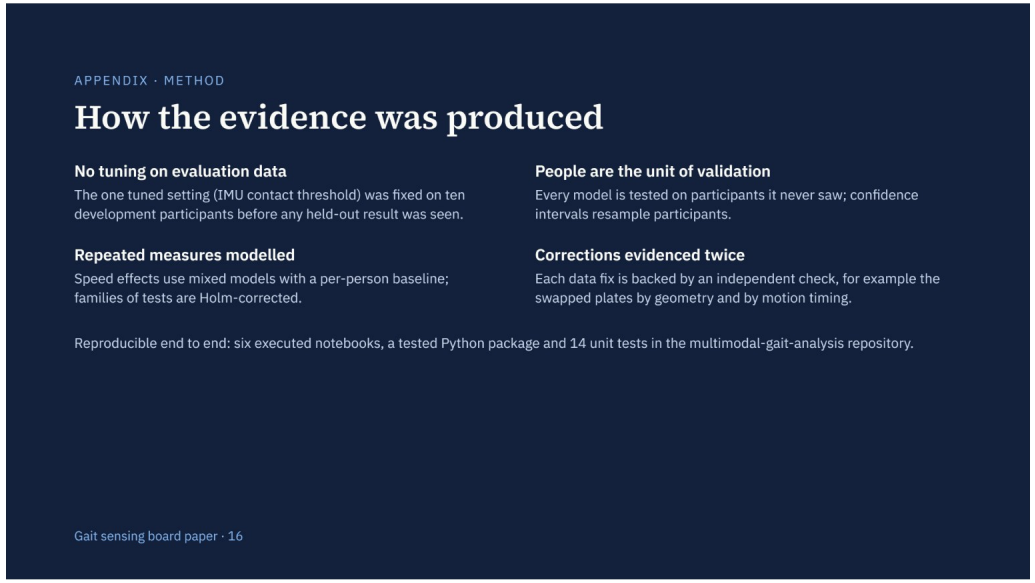
The gates turn this study's results into acceptance criteria. Phase 1 matters most: the evidence comes from healthy young adults on a treadmill, so the first spend should establish that the same accuracy holds for the people the service is for. Budgets and timelines are for the programme team to attach once the target population is fixed.

RISKS AND LIMITATIONS

What could make these conclusions wrong

Risk	Why it matters	Mitigation
Population	Healthy adults aged about 22, slow treadmill walking	Phase 1 re-validation in target users
Single session	Re-identification shown across speeds, not across days	Treat as biometric until shown otherwise
Inflated vendor claims	A naive data split overstated accuracy by 13 points here	Require participant-wise validation
Source data quality	20 issues in a published release, some silent	Audit every dataset before use
EEG hardware	Dry headset without motion reference channels	Gate EEG on purpose-built hardware
Processed EMG	Only a 16 Hz envelope released	Capture raw EMG in any new study

The largest risk is population transfer: nothing here shows the IMU performs as well for older or impaired walkers, which is why Phase 1 is a validation phase rather than a build phase.



For technical reviewers: the code regenerates every figure and number in this deck from the raw release in under ten minutes on a workstation.